PAK-AUSTRIA FACHHOCHSCHULE

Institute of Applied Sciences and Technology

School of Computing

ASSIGNMENT –3			
SPRING 2026			
Course Title	Intrusion Detection Systems	**Course Code**	COMP-995
Course Instructor	Dr Muhammad Zeeshan	**Program**	MS IS
Class		**Semester**	1,2
Submission Date	11 June 2026	Marks	100

Important:

- *This assignment refers to the topics covered in Lectures 17 onwards*
- *Students are encouraged to browse online resources and relevant research work*

1. Research Domains:

1. Cryptography
2. Cryptanalysis
3. Statistical characterization of cybersecurity datasets
4. Statistical Security Profile (SSP)
5. Artificial Intelligence

2. Core Theme

Cryptographic System Weakness Detection and Side-Channel Resilience Assessment using Statistical Profiling and AI-based Cryptanalysis

3. Assignment Overview

This assignment requires you to design and implement a research-grade framework that analyzes cryptographic system weaknesses using:

- Statistical Analysis (Statistical Security Profile – SSP)
- Machine Learning / Deep Learning
- Cryptanalysis techniques
- Side-channel analysis datasets

Task-1:

Understand Real Industry Problem & Challenge

Modern systems (IoT, embedded devices, cloud HSMs) rely on cryptography, but:

- Algorithms are theoretically secure but practically vulnerable
- Side-channel attacks (power, EM, timing) expose keys
- Weak randomness and poor implementations introduce leakage

Key Challenges

- No unified framework to:
    - Detect side-channel leakage
    - Analyze ciphertext statistical weaknesses
    - Evaluate randomness quality
- Existing tools:
    - Are manual or dataset-specific
    - Lack AI-driven automation
    - Lack statistical interpretability

Task-2:

Formulate Problem Statement

Develop a unified AI-driven framework that uses statistical profiling (SSP) and machine learning to detect cryptographic weaknesses and assess resilience against side-channel and cryptanalysis attacks.

Task-3:

Define Research Objectives

- Develop a Statistical Security Profile (SSP) for cryptographic datasets
- Detect side-channel leakage using AI models
- Analyze entropy and randomness of cryptographic outputs
- Identify statistical weaknesses in ciphertext distributions
- Build a multi-modal AI model (signal + statistical + ciphertext)
- Evaluate attack resilience of cryptographic systems
- Propose a Cryptographic Risk Index (CRI)

Task-4:

Relevant Dataset Selection

You MUST use datasets from at least two different categories:

Select one or more datasets from each category for your assignment

Side-Channel Datasets

- ASCAD Dataset

- DPA Contest Datasets
- ChipWhisperer Datasets

Cryptanalysis / Ciphertext

- CrypTool Ciphertext Corpus
- RockYou Dataset

Any other relevant dataset

IMPORTANT: Use multi-modal datasets together

Task-5:

Perform Randomness Evaluation:

For randomness evaluation, use one or more of the following metrics. Select randomness metrics aligned with your objectives.

1. NIST Randomness Test Suite Data
2. Information-Theoretic Metrics (Highly Important)
   a. Shannon Entropy
   b. Min-Entropy
   c. Conditional Entropy
   d. Mutual Information
3. Statistical Distribution Tests
   a. Frequency (Monobit) Test
   b. Block Frequency Test
   c. Runs Test
   d. Serial Test
   e. Longest Run of Ones Test
4. Pattern & Structure Detection Metrics
   a. Autocorrelation Function (ACF)
   b. Partial Autocorrelation (PACF)
   c. Hurst Exponent
5. Any other metric for randomness evaluation

Task-6:

Perform Statistical Analysis:

Objective of Statistical Analysis Module (SAM) is to perform deep statistical characterization before AI modeling. SAM is a critical pre-processing and analytical layer that performs deep statistical characterization of cybersecurity datasets before applying AI/ML models. Its main significance is to ensure that model development is data-aware, explainable, and scientifically grounded rather than purely algorithm-driven.

Processes:

A. Dataset-Level Statistics

- Mean, variance, std deviation

- Feature distributions

B. Entropy & Uncertainty Analysis

- Shannon entropy
- Conditional entropy
- Information gain

C. Correlation & Dependency Analysis

- Pearson correlation
- Mutual information
- Feature redundancy

D. Outlier & Anomaly Statistics

- Z-score
- IQR
- Outlier percentage

E. Class Imbalance Profile

- Class distribution
- Imbalance ratio
- Gini index

SSP Insight Examples

- Identify leakage-prone features
- Detect weak randomness
- Reveal structural bias in ciphertext

Task-7:

Write Statistical Security Profile (SSP) Report

The SAM produces a statistically enriched representation of cybersecurity datasets that bridges raw data and AI-driven intrusion detection models. It produces a set of structured analytical outputs that collectively form the Statistical Security Profile (SSP) and guide AI/ML modeling.

SSP is a comprehensive structured report containing dataset intelligence, including:

i) Dataset-Level Statistical Summary

- Mean, median, variance, standard deviation of features
- Min–max ranges

- Distribution shapes (skewness, kurtosis)
- Understanding of data spread and variability
- Detection of abnormal feature behavior

ii) Entropy & Uncertainty Profile

- Shannon entropy per feature
- Conditional entropy between features
- Information gain ranking
- Identifies predictable vs random features
- Highlights potential weak signals or strong indicators of attacks

iii) Correlation & Dependency Matrix

- Correlation heatmap (Pearson/Spearman)
- Mutual information scores
- Feature dependency graph
- Detects:
    - Redundant features
    - Highly dependent variables
    - Hidden relationships in traffic patterns

iv) Outlier & Anomaly Statistics Report

- Z-score based anomaly counts
- IQR-based outlier detection
- Percentage of abnormal samples
- Identifies irregular network behavior
- Helps detect early signs of attack traffic or noise

v) Class Imbalance Profile

- Attack class distribution table
- Imbalance ratio per class
- Gini index / entropy of labels
- Reveals bias in dataset
- Guides resampling strategies (SMOTE, weighting)

vi) Feature Importance Pre-Ranking (Statistical)

- Features ranked by:

    - variance
    - entropy
    - mutual information
- Pre-feature selection for ML models
- Reduces dimensionality before training

vii) Behavioral Traffic Profile (Cybersecurity Insight)

- Protocol distribution (TCP/UDP/ICMP)
- Flow duration patterns
- Packet size behavior trends
- Understands how attackers behave vs normal users
- Useful for traffic behavior modeling

viii) Visualization Artifacts

- Histograms of feature distributions
- Correlation heatmaps
- Entropy plots
- Boxplots (outliers)
- Time-series behavior graphs
- Human-interpretable dataset understanding
- Supports research paper figures

ix) Data Quality Assessment Report

- Missing value report
- Noise level estimation
- Data consistency score
- Ensures dataset is suitable for ML modeling
- Prevents biased or corrupted training

x) SSP-Based Insights Summary

- Most informative features for attack detection
- Highly correlated redundant features
- Attack patterns in statistical space
- Dataset reliability score

Task-8:

Formulate Methodology

Phase 1: Data Preprocessing and Statistical Profiling

- Normalize traces and ciphertext
- Align feature dimensions
- Handle imbalance
- Generate SSP
- Identify important features

Phase 2: Multi-Modal Statistical Fusion

It should combine Modality and Data

1. Signal (Power traces)
2. Statistical (SSP features)
3. Ciphertext (Encrypted data)

Phase 3: AI Model Design

A Hybrid Architecture is suitable for the model design:

- CNN → side-channel traces
- Transformer → ciphertext patterns
- Autoencoder → anomaly detection
- Fusion layer → final decision

Phase 4: Cryptanalysis Simulation

It should simulate:

- Differential Power Analysis (DPA)
- Entropy-based attacks
- Frequency analysis

Phase 5: Cryptographic Risk Index (CRI)

Define CRI, for example,

CRI = f(leakage score, entropy deviation, model confidence)

Task-9:

Performance Evaluation & Benchmarking

It should include:

Standard Metrics

- Accuracy

- Precision
- Recall
- F1-score
- ROC-AUC

Cryptography-Specific Metrics

- Key recovery rate
- Leakage detection rate
- Entropy deviation score

Efficiency Metrics

- Runtime
- Memory usage

Task-10:

Results Presentation

Students must include:

- SSP tables
- Entropy graphs
- Correlation heatmaps
- Model performance graphs
- Cross-dataset results

Task-11:

Comparison with State-of-the-Art

Classical Methods

- Differential Power Analysis (DPA)
- Correlation Power Analysis (CPA)

Task-12:

Discussion & Analysis

Interpret:

- Why certain features leak information
- How entropy correlates with attack success
- Model strengths/limitations
- Practical implications

AI-Based Methods

- CNN-based SCA
- LSTM-based models

Comparative Dimensions

- Accuracy
- Key recovery success
- Generalization
- Interpretability

Task-13:

Conclusion

- Summary of findings
- SSP + AI improves cryptanalysis
- Demonstrated unified framework
- Contributions to research

Task-14:

Technical Report

Technical report consists of description of Task-1 to Task-13, organized in 13 sections, additionally, title, abstract and Introduction. Results section must include results in the form of tables and graphs.

13. Marking (Total = 100 Marks)

Task #	Marks
Task-1	5
Task-2	5
Task-3	5
Task-4	5
Task-5	10
Task-6	10
Task-7	10
Task-8	15
Task-9	10

Task-10	10
Task-11	5
Task-12	5
Task-13	5
Total	**100**

14. Bonus (Up to +10 Marks)

- Multi-modal AI model
- Explainable AI (SHAP/LIME)
- Cross-dataset evaluation
- Novel CRI formulation

Note:

i. Assignment submission must include:
   a. Technical Report in MS WORD format
   b. The source code
ii. Assignments will be uploaded in MS Teams class group
iii. 50% marks will be deducted in case of late assignment submission
iv. *No copying of other student's work is allowed. Zero marks will be awarded in case of cheating*